

FlowSight

A Real-Time Data Engineering Pipeline for Traffic Analytics

Team:7
Members :

S.NO	NAME	ROLL NUMBER
1	J.NANDINI	2420030262
2	S.SAI SARANYA	2420030266
3	B.SAI CHARNA	2420030364

Faculty Guide : Dr. N Sirisha

(Associate professor)

Overview

01 **Abstract**

02 **Introduction**

Abstract

FlowSight: A Real-Time Data Engineering Pipeline for Traffic Analytics is a Data Engineering project designed to collect, stream, process, store, and analyze continuously generated traffic data. The system uses a Python-based traffic sensor simulator to generate realistic traffic events containing sensor ID, road name, timestamp, vehicle count, and average speed. These events are **streamed in real time** using **Apache Kafka**, which acts as the **data ingestion and streaming layer**. The incoming data is validated to identify missing, duplicate, or invalid records and is then transformed into a consistent, analytics-ready format. Additional information such as hour, day, weekday, and congestion level can be derived from the raw traffic readings. This approach demonstrates how raw and continuously arriving data can be converted into reliable structured information through an automated data pipeline.

The **processed traffic data** is stored in **PostgreSQL** for persistent and historical analysis. **SQL** is used to **calculate important traffic metrics** such as **vehicle volume**, average speed, peak traffic hours, and congested roads. The **results can then be presented** through **Power BI dashboards**, allowing users to understand traffic patterns and compare road conditions easily. The main objective of FlowSight is to demonstrate the **Fundamentals of Data Engineering** lifecycle, including **data generation, real-time ingestion, streaming, data validation, transformation, storage, aggregation, analytics, and visualization**. The project provides a practical example of how Data Engineering technologies can work together to transform continuous raw traffic data into meaningful, trustworthy, and analytics-ready information.

Introduction

In today's data-driven environment, cities and transportation systems can generate large volumes of traffic information from sensors and monitoring systems. Traffic data may include vehicle counts, average speed, road identifiers, timestamps, and other operational measurements. As the number of sensors and frequency of readings increase, raw traffic data becomes difficult to manage and analyze manually.

FlowSight automates collection, streaming, validation, transformation and storage of traffic data, then converts it into useful analytics for congestion and traffic-pattern analysis.

Data Engineering provides the processes and technologies required to collect, ingest, validate, transform, store, and organize this data into reliable information. FlowSight applies these principles to a traffic analytics scenario. A Python simulator represents virtual traffic sensors, Apache Kafka provides real-time streaming, PostgreSQL provides persistent storage, and SQL supports analytical queries.

The project focuses on building the data pipeline rather than directly controlling traffic signals or predicting traffic using machine learning. Its purpose is to demonstrate how continuous raw events can be converted into clean, structured, analytics-ready data that can support understanding of congestion patterns, peak hours, road performance, and historical traffic trends.

Thank You
